

Abhishek K. Dubey

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Profile Summary

Strategic R&D leader with 17+ years of experience fusing advanced statistical methodologies with AI/ML innovation — from foundation models for digital pathology to adaptive clinical trial design — in service of precision oncology drug development. At Bristol Myers Squibb, I lead statistical strategy for antibody-drug-conjugate and combination portfolios spanning early design through BLA submission, translating multimodal clinical and molecular data into registrational evidence. I bring a consistent record of cross-functional leadership, regulatory engagement, and delivery of high-impact analytical strategies, executed with decisive accountability and urgency in highly regulated environments.

Core Competencies

Clinical Trial Design & Statistics:	Adaptive & MAMS designs, dose-escalation/dose-optimization frameworks, integrated benefit-risk assessment, external-control & data-borrowing methods, Probability of Technical Success (PoTS) modeling, time-to-event analysis
Regulatory & Data Standards:	Protocol & SAP authorship, DMC charters, ADaM/SDTM data review, IND/BLA submission strategy, health authority interactions
AI/ML & Computational Methods:	Deep learning, weakly supervised multitask learning, generative AI/foundation models for digital pathology, domain adaptation, uncertainty quantification, medical image registration
Leadership:	Cross-functional program leadership (Clinical, Regulatory, Commercial), portfolio-level statistical strategy, mentoring & team development

Education

Duke University <i>Doctor of Philosophy (Ph.D.) in Computer Science</i>	08/20/2012 – 09/17/2018 Durham, NC
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Indian Institute of Technology (IIT) Kharagpur <i>Bachelor of Technology (B.Tech.) in Computer Science (Top 0.1% nationwide)</i>	08/12/2005 – 05/01/2009 Kharagpur, India
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Industry & Academic Experience

Bristol-Myers Squibb <i>Associate Director, Biostatistics, Data & Quantitative Sciences</i> <i>Senior Manager, Biostatistics, Data & Quantitative Sciences</i>	Lawrence Township, NJ 06/15/2026 – Current 12/06/2021 – 06/15/2026
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- Indication Lead directing statistical vision for ADC and combination portfolios across BC, HNSCC, and pediatrics, balancing direct study execution with macro-portfolio oversight.
- Provided strategic statistical leadership for 6+ high-priority Phase 1–3 programs across NSCLC, TNBC, HCC, and Melanoma, accelerating timelines to registrational pathways.
- Partnered with Cross-Functional Leadership (Clinical, Regulatory, and Commercial) to architect end-to-end global development strategies from early design through BLA submission.
- Directed program-level pooled safety strategy across 15+ studies, delivering data-driven toxicity insights that shaped portfolio-wide management and regulatory labeling strategies.
- Pioneered advanced adaptive trial frameworks, designing a pivotal Phase 2/3 multi-arm multi-stage (MAMS) registrational program featuring adaptive dose-dropping strategies and strategic borrowing methodologies.
- Optimized development pathways through quantitative assessments of master versus independent protocols, maximizing design efficiency and commercial value within an evolving competitive landscape.
- Authored core regulatory statistical deliverables — protocols, SAPs, DMC charters, and CSR data-presentation plans — for programs advancing towards registration.
- Led high-stakes statistical interactions with health authorities on trial design and regulatory strategy.

- Established and led the program’s inaugural Data Monitoring Committee in ADC and combination portfolios. Coordinated interim analyses and independent statistical centers supporting DMC oversight.
- Developed integrated benefit-risk assessment frameworks to inform dose-selection decisions.
- Designed non-concurrent data-borrowing methodology for contribution-of-component evaluation.
- Architected model-augmented external-control framework to strengthen single-arm and hybrid trial designs.
- Architected a modular machine learning framework to generate counterfactual benchmarks, quantifying therapeutic synergy versus redundant efficacy in drug combinations.
- Executed Probability of Technical Success modeling to quantify risk, directly informing high-value portfolio investment and commercial positioning strategy.
- Spearheaded statistical components of successful IND and BLA submissions.
- Supervised programming teams through rigorous ADaM and SDTM data reviews.
- Mentored 8+ Biostatisticians and interns
- Co-led GenAI integration utilizing digital H&E images to predict high-risk patient status and expedite enrollment for registrational programs.

National Cancer Institute (NCI)

12/15/2020 – 12/05/2021

Research Fellow, Cancer Data Science Laboratory, Center for Cancer Research

Bethesda, MD

- Led development of a foundational H&E AI platform (Multitask-TwoScale-HENet) to convert routine pathology slides into predictive assets for survival, molecular characterization, immune profiling, and clinical risk stratification.
- Established a scalable weakly supervised multitask-instance framework enabling biomarker inference and patient enrichment without costly pixel-level annotation.
- Demonstrated translational utility through external validation, survival-based risk stratification, and tumor microenvironment characterization aligned with biological literature.
- Integrated statistical rigor and governance safeguards (interpretability, robustness testing, domain-shift monitoring) to support responsible deployment in drug development settings.
- Positioned H&E-derived modeling as a strategy to bridge biomarker data gaps in early-phase oncology programs and improve probability of technical success.

Oak Ridge National Laboratory (ORNL)

11/19/2018 – 11/16/2020

Postdoctoral Researcher Associate, Advance Computing for Health Sciences

Oak Ridge, TN

- Led end-to-end productization of an AI-powered information extraction platform for oncology pathology reports supporting SEER registry operations, converting unstructured clinical text into structured, regulatory-grade real-world data.
- Partnered with clinicians, cancer registrars, data scientists, and engineering teams to translate research prototypes into deployable, scalable solutions aligned with operational workflows.
- Implemented probabilistic deep learning models with uncertainty quantification to ensure reliability, auditability, and stakeholder trust in production environments.
- Drove domain-adaptation strategies to improve generalizability of chest radiograph abnormality detection across heterogeneous clinical sites.
- Evaluated system-level performance trade-offs (e.g., image resolution vs. accuracy vs. compute cost) to inform scalable deployment decisions in healthcare AI workflows.

Duke University

08/20/2012 – 09/17/2018

Graduate Research Associate (Prof. Xiaobai Sun; Prof. Shivnath Babu)

Durham, NC

- Developed mathematically grounded frameworks for deformable medical image registration, addressing computational and accuracy bottlenecks in CT/MRI alignment.
- Introduced a novel bi-residual inversion and feedback-control methodology for accurate and efficient deformation vector field (DVF) inversion with theoretical guarantees.
- Designed a two-step forward-backward registration framework eliminating traditional computational overhead while preserving symmetry and stability.
- Enabled clinically viable solutions for precision radiation dose planning and automated atlas-based segmentation, advancing reliability in image-guided precision medicine.

IBM Research

05/10/2014 – 08/10/2014

Research Intern

Almaden, CA

- Self-adaptive NoSQL stack — a storage workload perspective.

Oracle India

Member Technical Staff, Server Technology and Database Management

- Developed user interface and backend solutions for enterprise solutions.

07/17/2009 – 06/01/2012

Bangalore, India

EPFL

Undergraduate Research Intern (Prof. Viktor Kuncak)

- Semantic representation of controlled English in extended discourse representation.

05/05/2008 – 08/05/2008

Lausanne, Switzerland

Honors & Awards

- **OLCF Director's Discretion Project Allocation & XSEDE Startup Allocation (NCI)**, 2020–2021 – Competitive national HPC awards supporting large-scale biomedical AI research.
- **Best Teaching Assistant Award**, Duke University, 2015–2016 (Numerical Analysis); Nominee, Probabilistic Machine Learning.
- **Graduate School Fellowship**, Duke University, 2012–2014.
- **Finalist, Best B.Tech Dissertation**, IIT Kharagpur, 2008–2009; **Merit Scholarship**, IIT Kharagpur, 2005–2009.

Selected Publications

- S Chakraborty, **Abhishek K Dubey**, et al. “A two-stage dose optimization framework for oncology drug development with multiple efficacy and safety endpoints via an integrated benefit-risk assessment.” *Statistics in Biopharmaceutical Research*, 2026.
- Md Inzamam Ul Haque, **Abhishek K Dubey**, et al. “The Effect of Image Resolution on Automated Classification of Chest X-rays.” *Journal of Medical Imaging*, 2023.
- A Ziabari, S Venkatakrishnan, **Abhishek K Dubey**, et al. “SIMURGH: A Framework for CAD-Driven Deep Learning Based X-ray CT Reconstruction.” *IEEE ICIP*, 2022.
- Arjun Chaudhuri, Ching-Yuan Chen, Jonti Talukdar, Siddarth Madala, **Abhishek K Dubey**, Krishnendu Chakraborty. “Efficient Fault-Criticality Analysis for AI Accelerators using a Neural Twin.” *IEEE ITC*, 2021.
- **Abhishek K Dubey**, Todd Young, Chris Stanley, Dalton Lunga, Jacob Hinkle. “Computer-aided abnormality detection in chest radiographs in a clinical setting via domain-adaptation.” *Bioimaging*, 2021.
- Shalin Shah, **Abhishek K Dubey**, John Reif. “ExTemp: A computational and image-processing suite for extracting temporal barcodes.” *JOSS*, 2020.
- Devanshu Agrawal, **Abhishek K Dubey**, Georgia Tourassi, Jacob Hinkle. “Deep kernel learning for information extraction from cancer pathology reports.” *CAFCW*, 2019.
- **Abhishek K Dubey**, Jacob Hinkle, James B Christian, Georgia Tourassi. “Extraction of Tumor Site from Cancer Pathology Reports using Deep Filters.” *ACM-BCB*, 2019.
- A.S. Iliopoulos, **A. Dubey**, and X. Sun. “idvf: Iterative inversion of deformation vector field with adaptive bi-residual feedback control.” *JOSS*, 2019.
- **Abhishek K Dubey**, Hong-Jun Yoon, Georgia Tourassi. “Localized Sliced Inverse Regression for Tumor Site Extraction From Cancer Pathology Reports.” *IEEE-BHI*, 2019.
- Shalin Shah, **Abhishek K Dubey**, John Reif. “Improved optical multiplexing with temporal DNA barcodes.” *ACS Synthetic Biology*, 2019.
- Shalin Shah, **Abhishek K Dubey**, John Reif. “Programming temporal DNA barcodes for single-molecule fingerprinting.” 2019.
- **Abhishek K Dubey**, A.S. Iliopoulos, X. Sun, F.F. Yin. “Iterative Inversion of Deformation Vector Fields with Feedback Control.” *Medical Physics*, 2018.
- B. Mitra, **Abhishek K Dubey**, S. Ghosh, and N. Ganguly. “How do Superpeer Networks Emerge?” *IEEE INFOCOM*, 2010.
- B. Mitra, **Abhishek K Dubey**, S. Ghosh, and N. Ganguly. “Formal understanding of emergence of super-peer in peer to peer Network.” *ICDCN*, 2010.

Selected Abstracts & Oral Presentations

- S Loi, **Abhishek K Dubey**, et al. “Trial in progress: A phase 2/3 trial of iza-bren (BMS986507/BL-B01D1), an EGFRxHER3 antibody-drug conjugate, vs standard-of-care chemotherapy in patients with previously untreated, locally advanced, recurrent inoperable, or metastatic triple negative breast cancer ineligible for anti-PD-(L)1 treatment (IZABRIGHT-Breast01).” *Clinical Cancer Research*, 2026.
- **Abhishek K Dubey** et al. Model-augmented external control for evaluating an experimental immuno-oncology agent. *IISA*, 2023.
- **Abhishek K Dubey**, Kaushal Mishra, Arun Kumar. “Improving dose-escalation design with historical and concurrent trial data.” *Joint Statistical Meetings*, 2022.
- Amir Ziabari, **Abhishek K Dubey**, et al. “High-Resolution X-Ray CT Reconstruction of Additively Manufactured Parts using GAN-based Unsupervised Domain Adaptation in AI-CT.” *Microscopy and Microanalysis*, 2021.
- Amir Ziabari, **Abhishek K Dubey**, et al. Additive Manufacturing: Advanced Characterization for Industrial Applications, 2021.
- Amir Ziabari, **Abhishek K Dubey**, et al. “Synthesizing X-Ray Computed Tomography Reconstruction Twins Using Semi-Supervised GANs.” *IMECE*, 2021.

- **Abhishek K Dubey**, et al. “Incisive Inversion of Deformation via Bi-residual Feedback.” CMCIM, 03/22/2020.
- **Abhishek K Dubey**, et al. AI in biomedical research. Invited webinars at Michigan, Stanford, Harvard, Dayton, and MD Anderson (2020).
- Shalin Shah, **Abhishek K Dubey**, John Reif. “Temporal DNA barcodes for single molecule imaging.” FNANO, 2019.
- Abnormality detection in chest radiographs. Bioimaging 2021.
- Extraction of Tumor Site from Cancer Pathology Reports. ACM-BCB, Niagara Falls, NY (09/07/2019).
- Localized Sliced Inverse Regression for Tumor Site Extraction. IEEE-BHI, Chicago, IL (05/19/2019).
- **Abhishek K Dubey**. Inversion of Small and Large Deformations. Invited seminars at Monsanto, ORNL, Duke Medicine, 2018.
- **Abhishek K Dubey**. Iterative Inversion of Deformation Vector Fields. AAPM, 2017.
- Self-adaptive NoSQL stack. IBM Research, Almaden, CA (08/02/2014).

Patents and Disclosures

System and method for artifact reduction of x-ray CT reconstruction leveraging AI and a priori known model for the object of interest. U.S. Nonprovisional Application No. 17/392,645.

References

References available upon request.